Ethanol Blending in Gasoline – Croatia

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

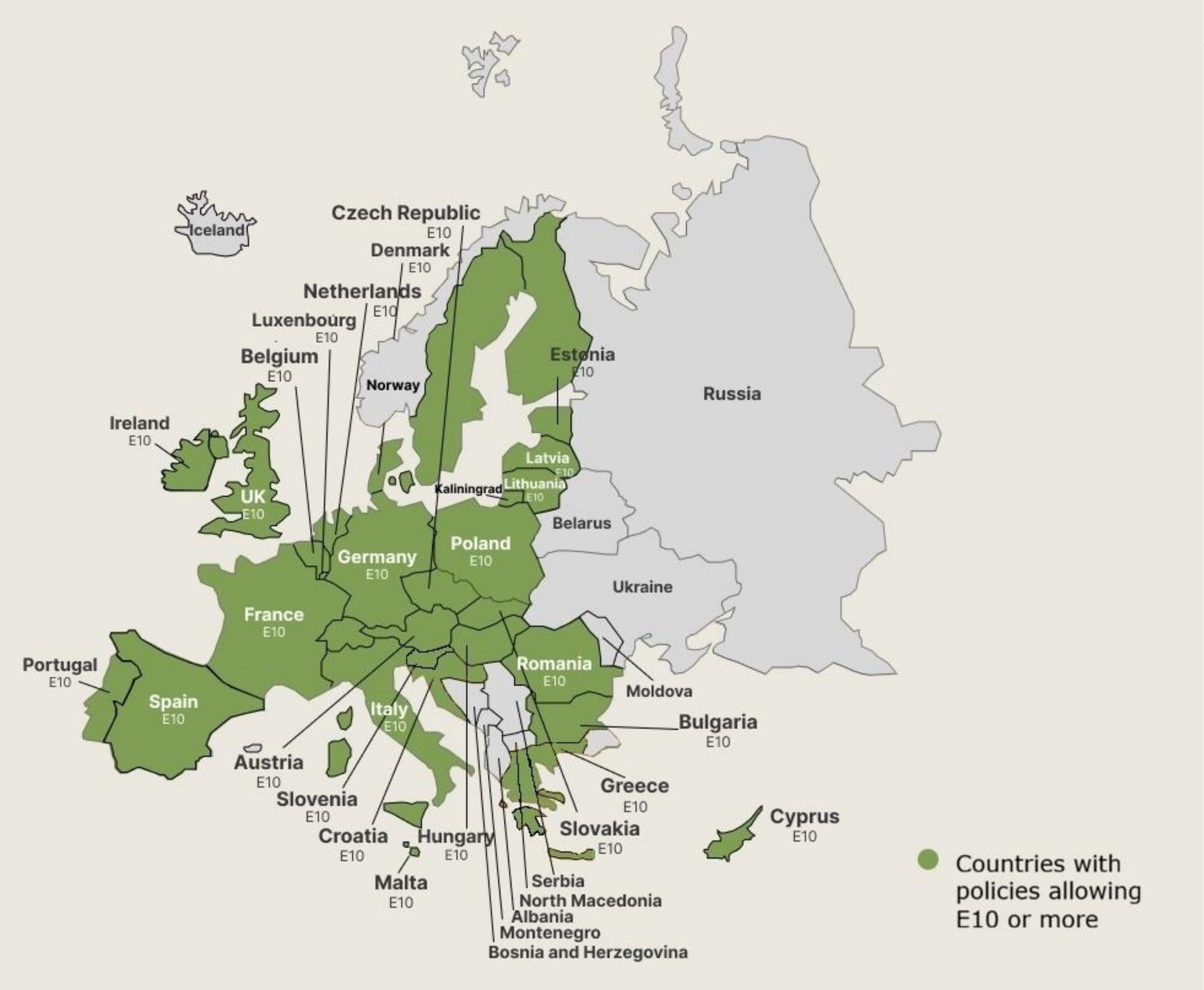
FARO90

www.faro90.com
+52 55 6122 2255

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

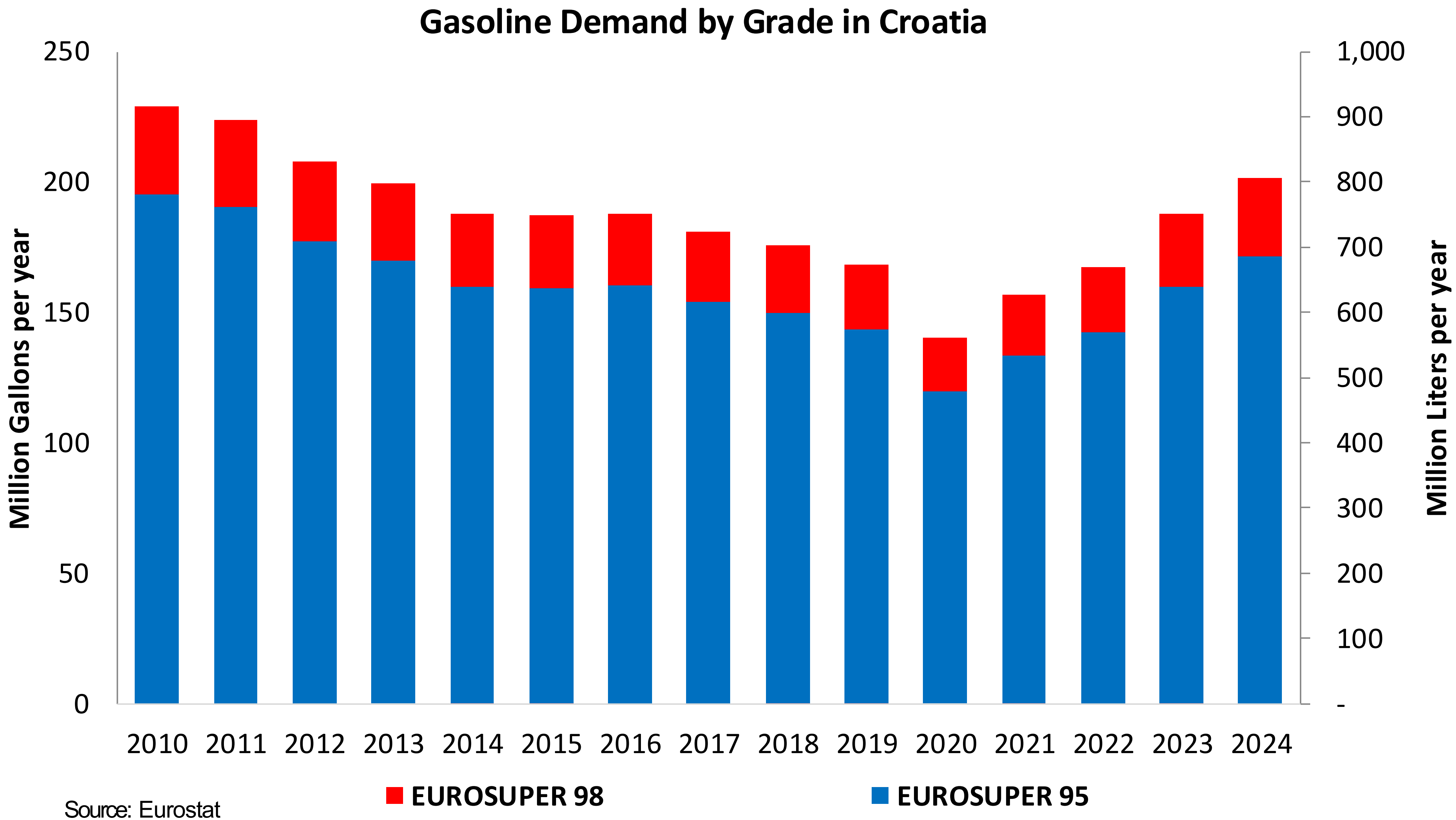
Ethanol Gasoline Blending in Croatia

In 2024, gasoline consumption in Croatia was 763 million liters (202 million gallons) and growth rate was 3.7%. Gasoline production and net imports were 709 and 82 million liters (187 and 22 million gallons) correspondingly. Croatia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

Altough E10 is allowed in Croatia, it does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

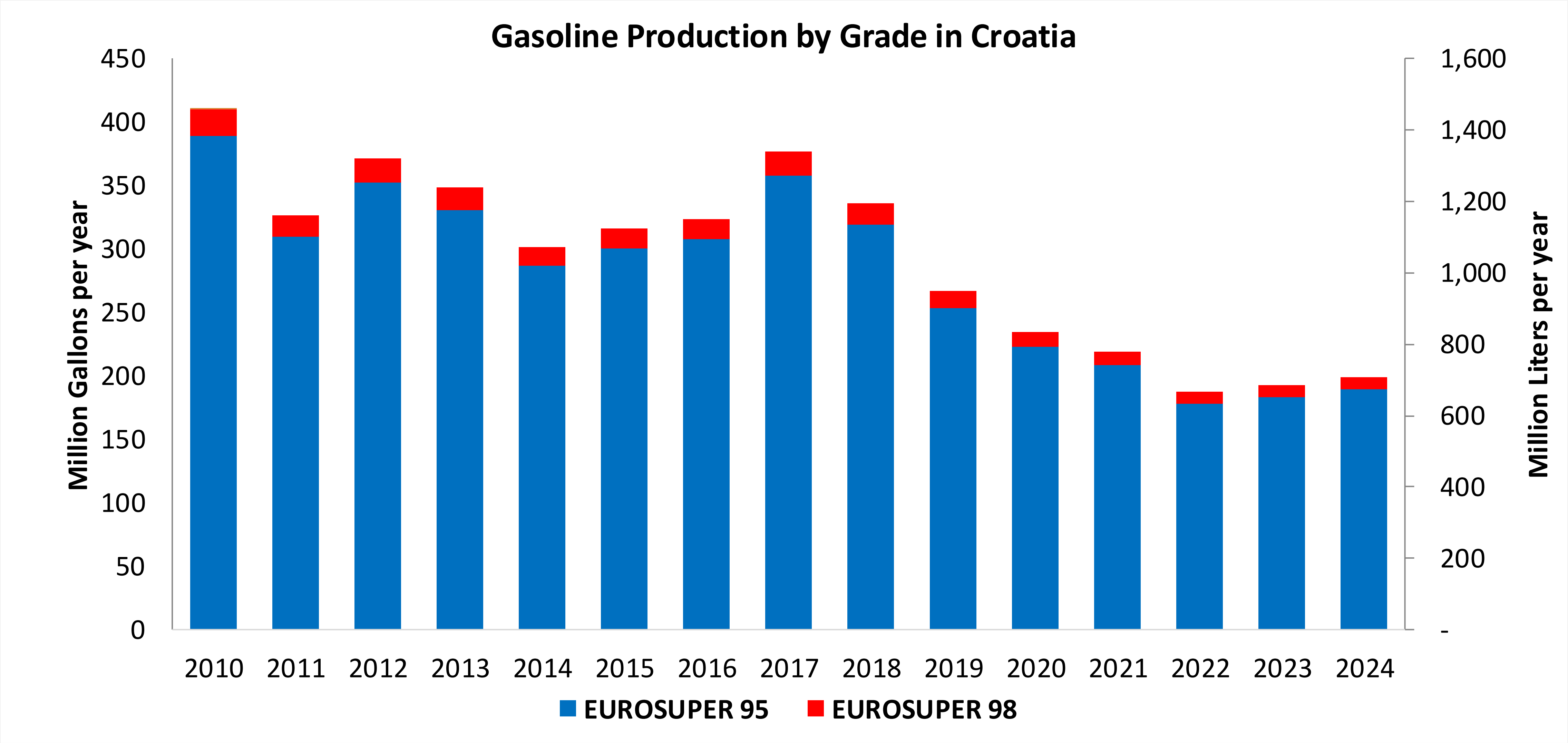
Source: Eurostat, European Commision

Gasoline Demand in Croatia



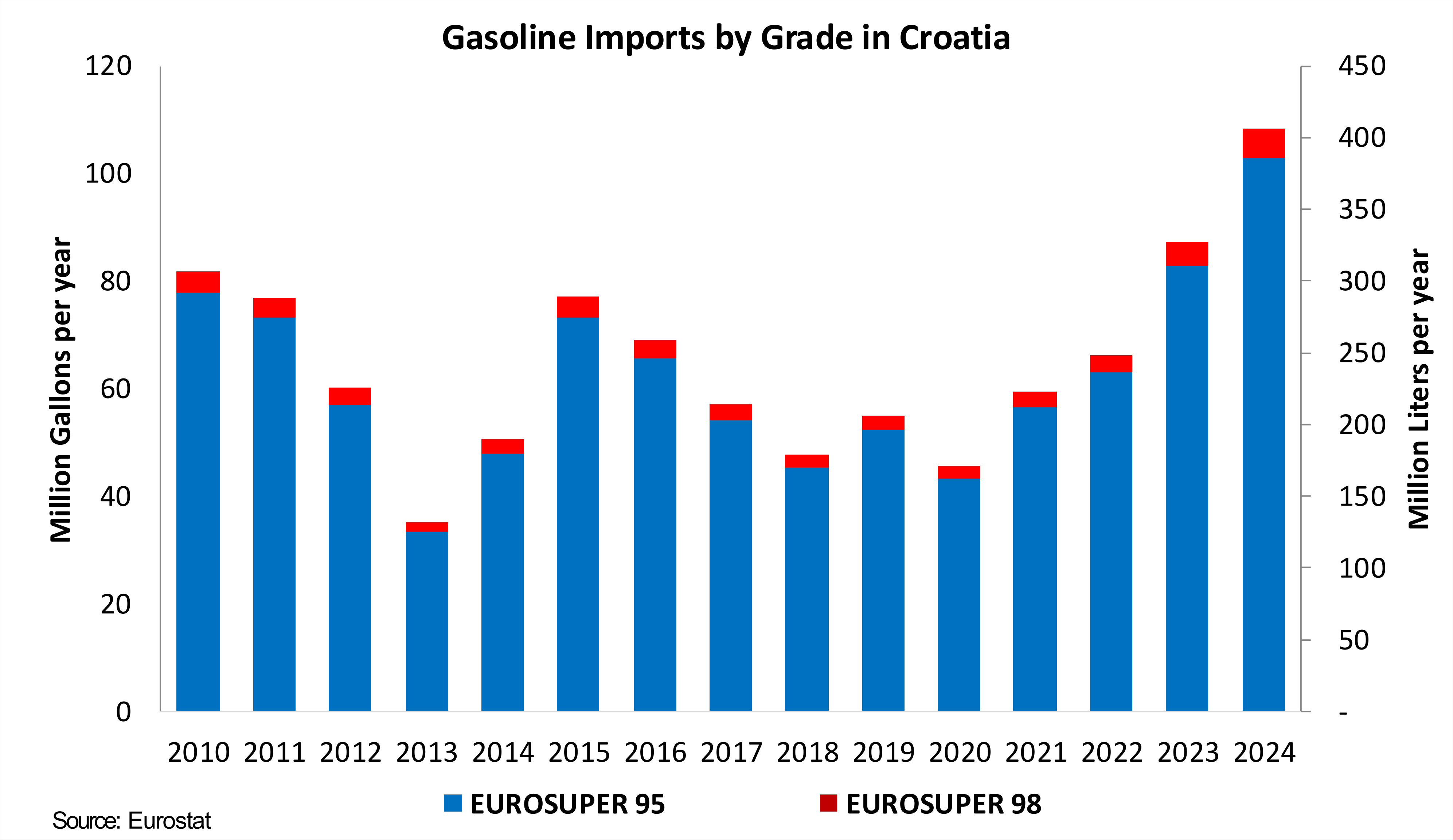
Source: Eurostat

Gasoline Production in Croatia

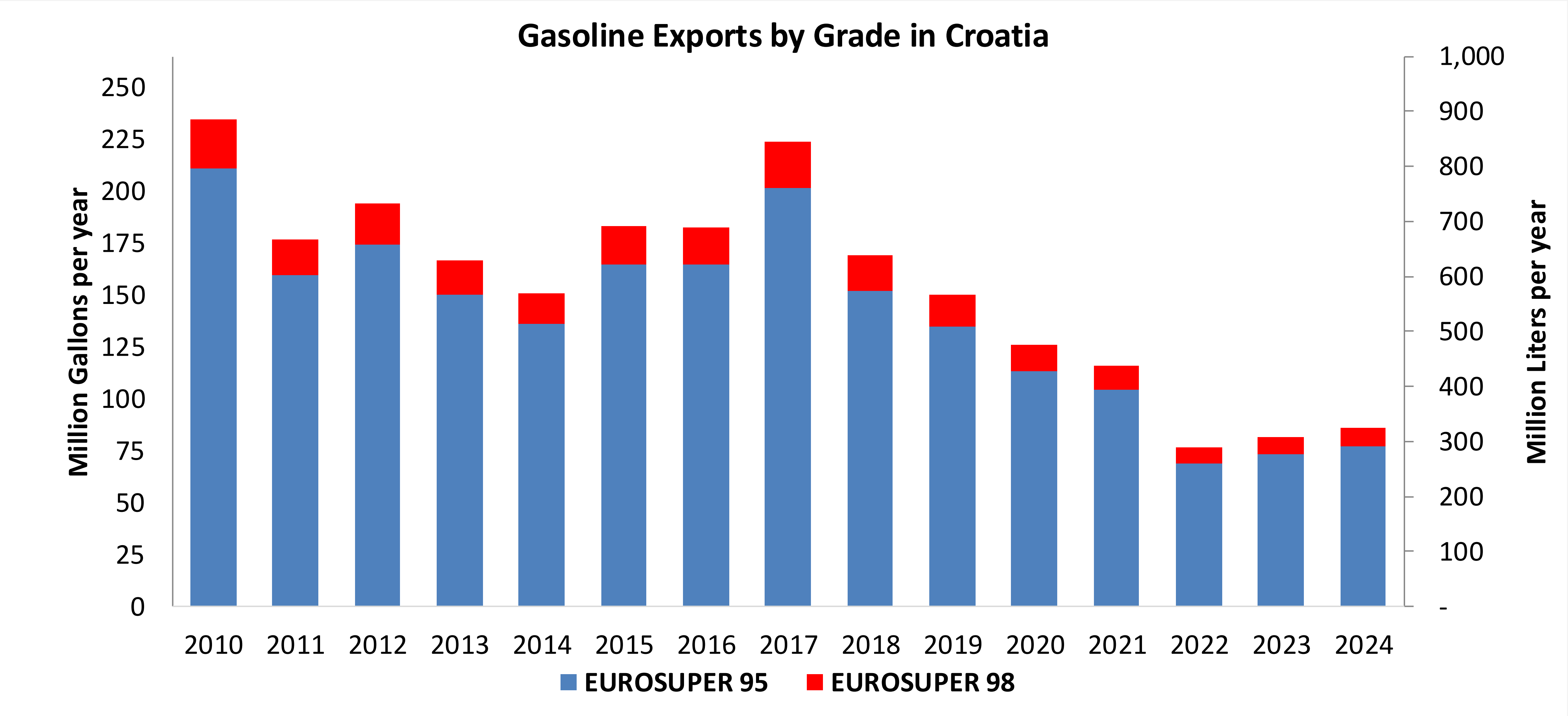


Source: Eurostat

Gasoline Imports in Croatia



Gasoline Exports in Croatia



Source: Eurostat

Ethanol Balance in Croatia

Croatia does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

Source: The Global Economy, WITS/Comtrade, Total Croatia News, Axens

Gasoline Quality in Croatia

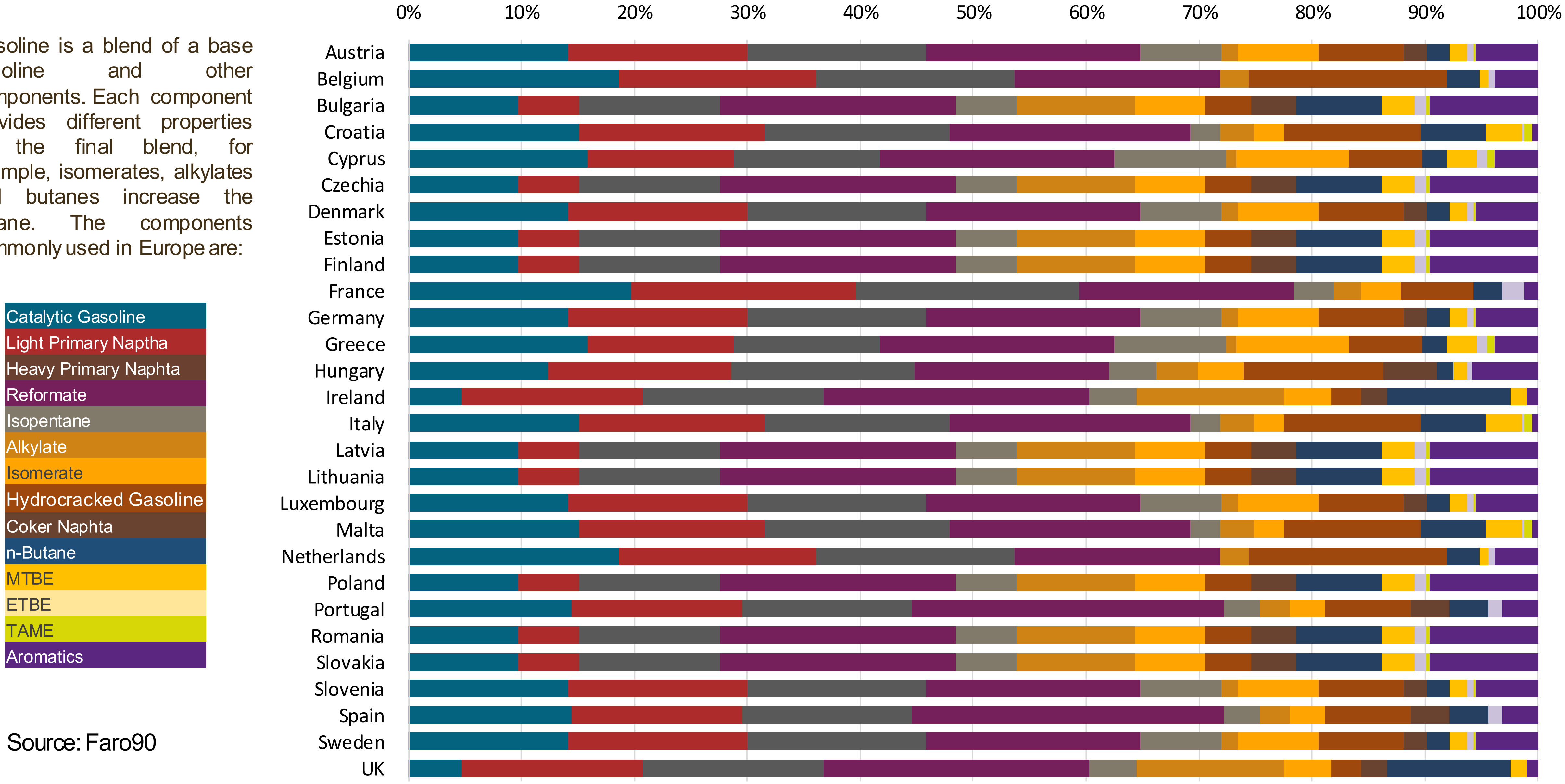
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.8							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	1.0							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

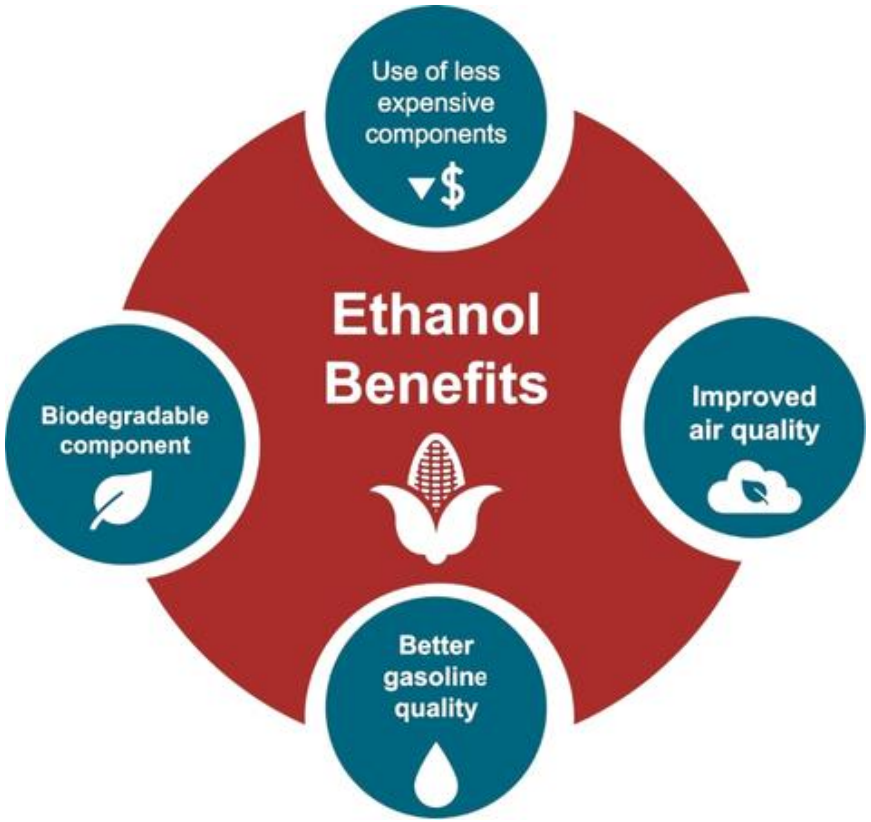
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

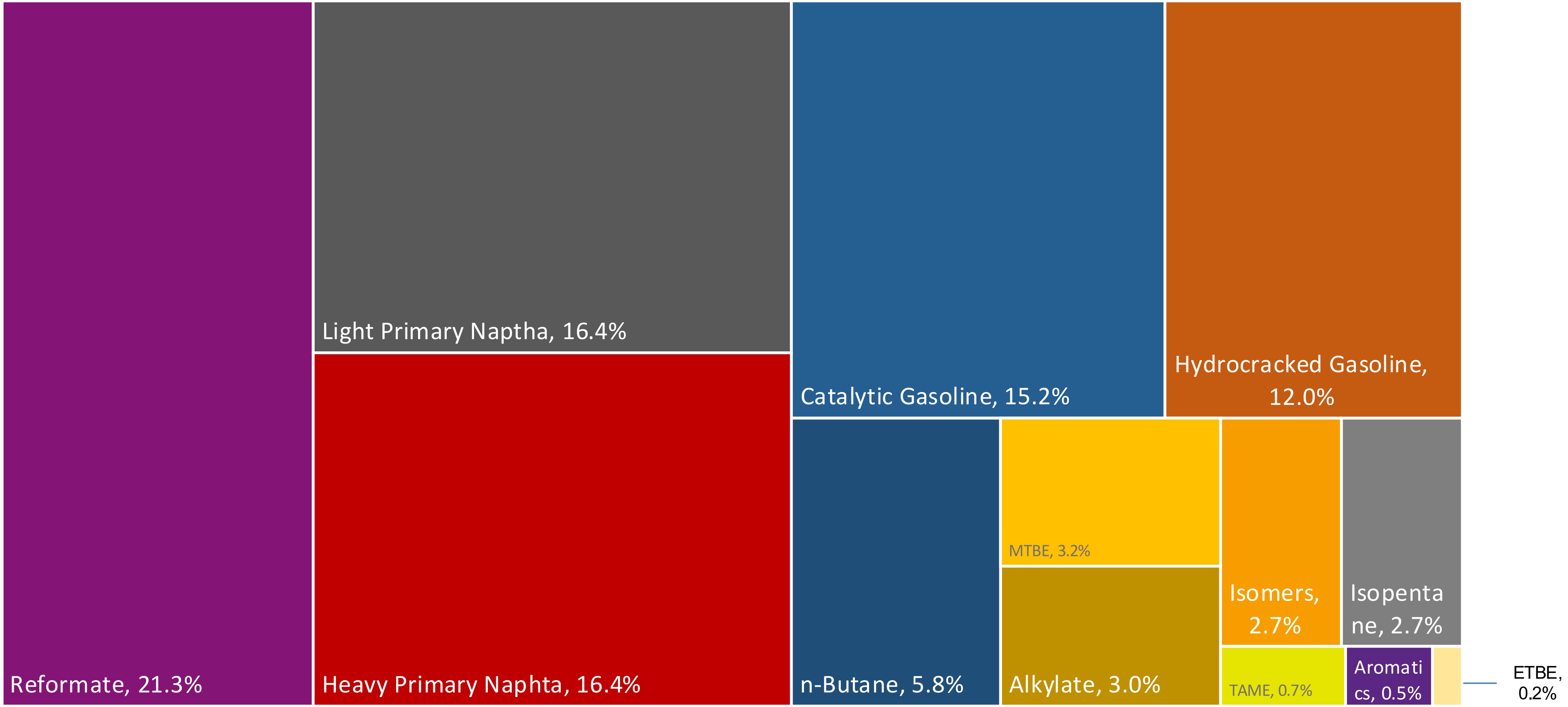
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

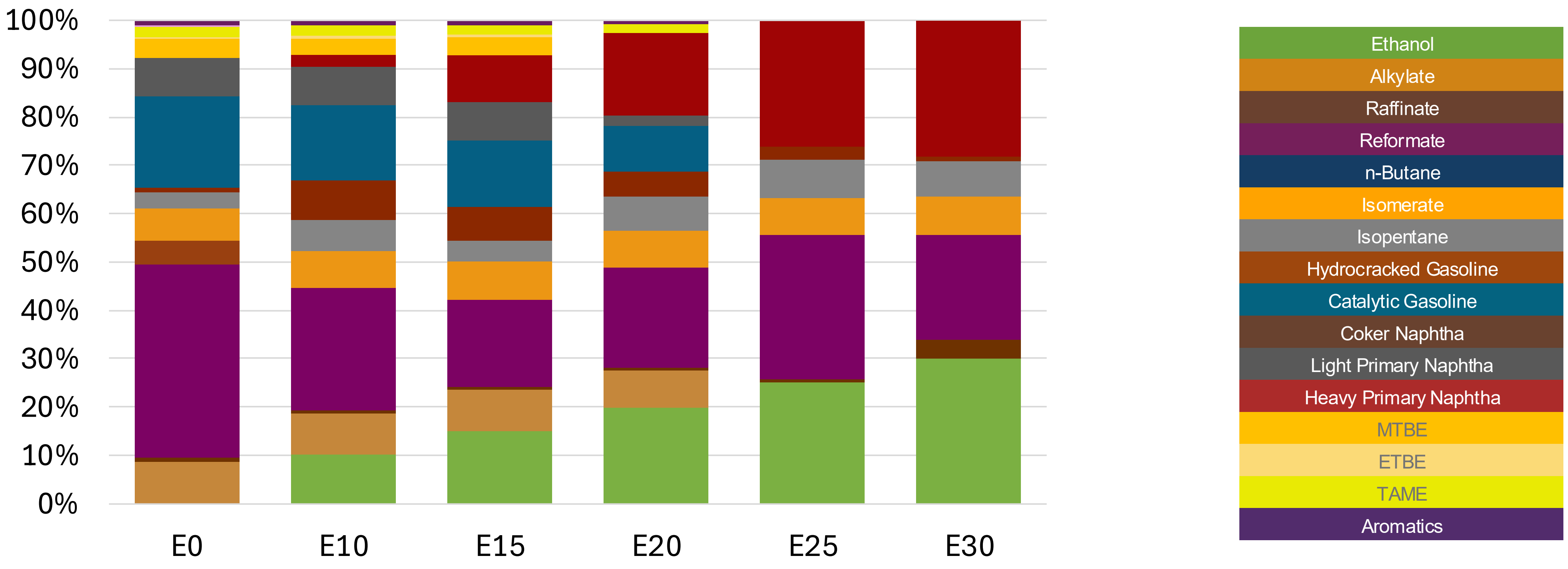


Croatia Available Blending Components



Source: Faro90

Croatia – Gasoline 95 – Constant Octane

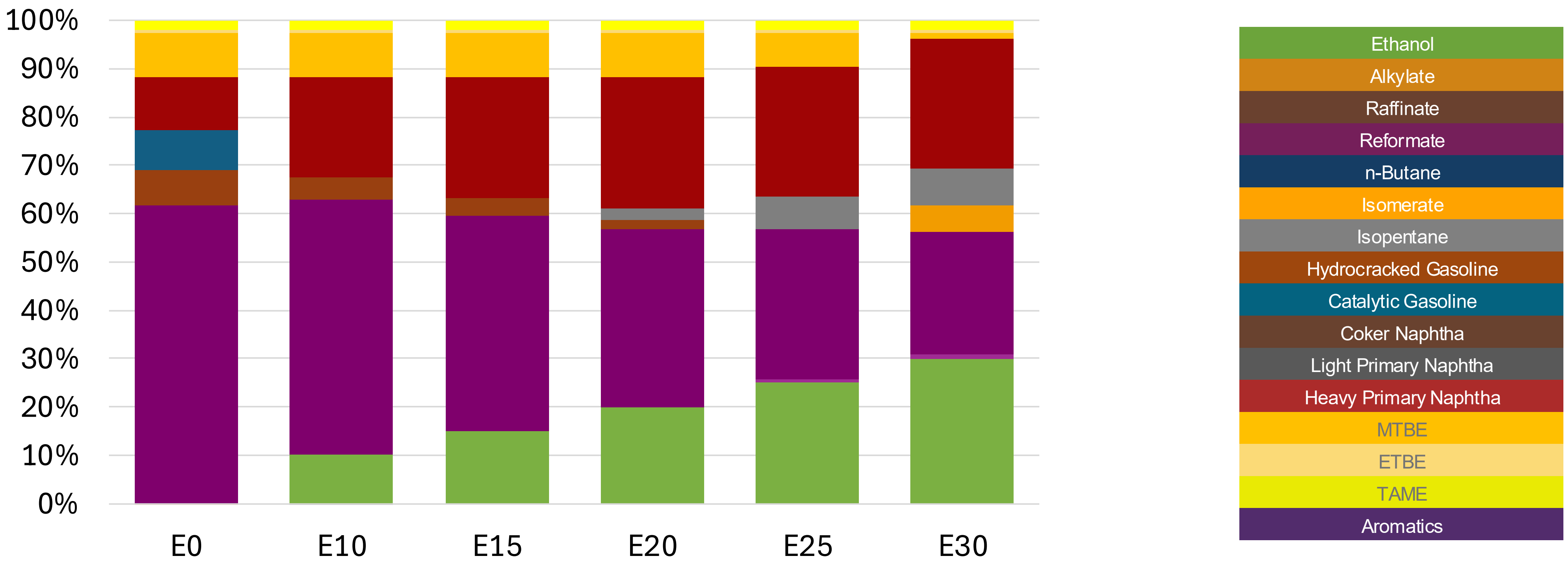


Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US$/gal)	$ 2.258	$ 2.145	$ 2.062	$ 1.986	$ 1.891	$ 1.817

Average CIF 2024 Prices

Source: Faro90

Croatia - Gasoline 98 - Constant Octane

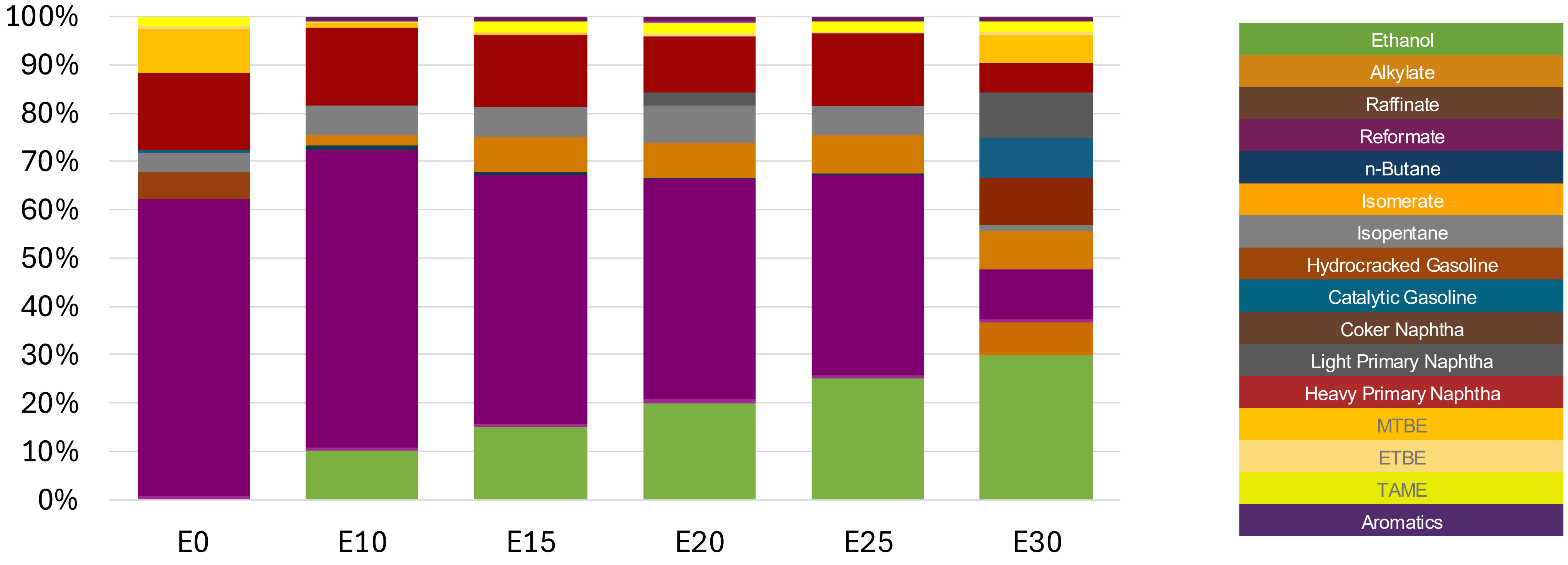


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.307	$ 2.188	$ 2.119	$ 2.049	$ 1.979	$ 1.926

Average CIF 2024 Prices

Source: Faro90

Croatia – Gasoline 95 – Increased Octane

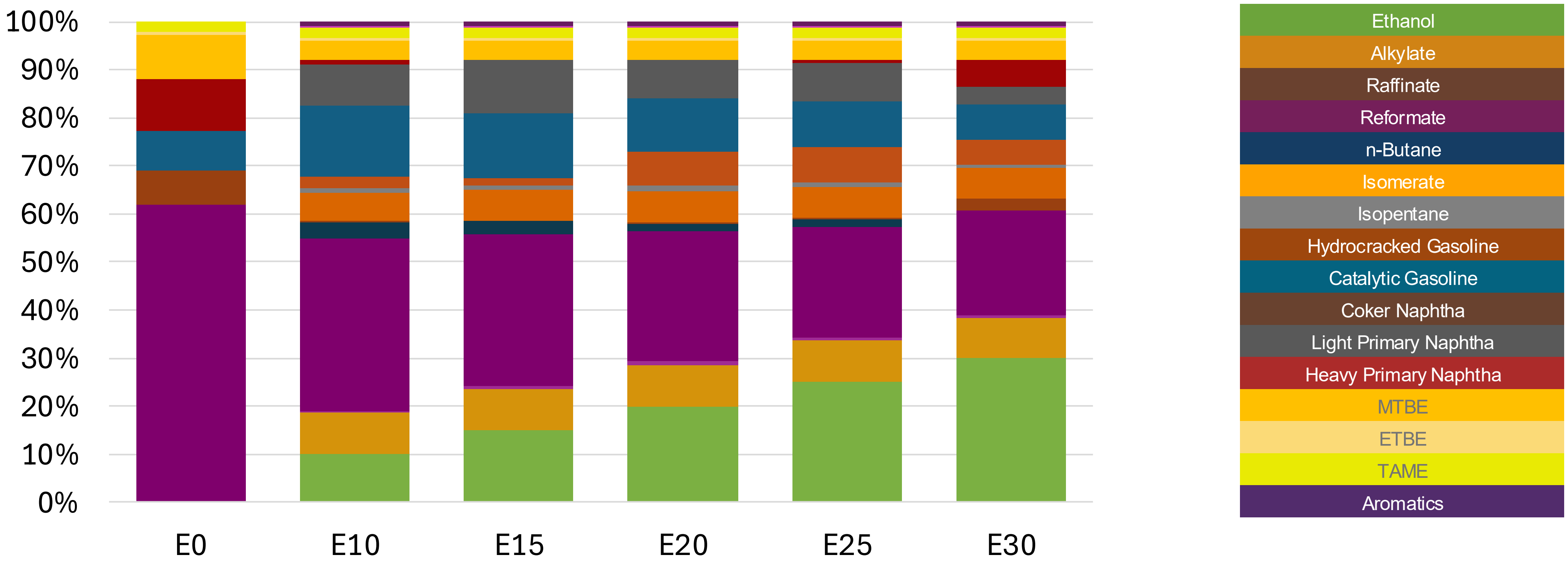


Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US$/gal)	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177

Average CIF 2024 Prices

Source: Faro90

Croatia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US$/gal)	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

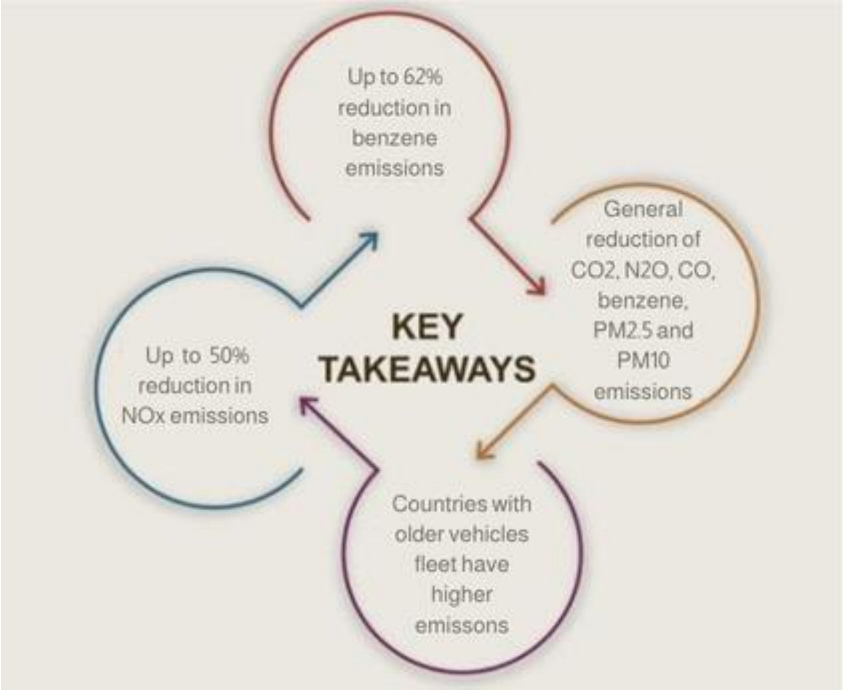
The model used in this analysis takes as a reference the [International Vehicle Emissions Model (IVE).]
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Croatia

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	634
		Single FI	3-Ways		>8 years	25,598
		Multi FI	3-Ways		>8 years	175,214
			Euro III		>8 years	
			Euro IV		>8 years	
			Euro V		>8 years	40,031
			Euro 6		>8 years	65,949
					4-8 years	120,875
					<4 years	45,411
			Hybrid		All	20,269
		Electric			All	3,283
	Medium	Carburator	3-Ways	PCV Tank	>8 years	63
		Single FI	3-Ways		>8 years	727
		Multi FI	3-Ways		>8 years	26,413
			Euro III		>8 years	42,880
			Euro IV		>8 years	89,459
			Euro V		>8 years	12,243
			Euro 6		>8 years	21,983
					4-8 years	55,443
					<4 years	35,708
			Hybrid		All	17,057
		Electric			All	2,352
Sml Engines	Light	4-Cycle	None	PCV	>8 years	30,933
			Euro III		>8 years	37,503
			Euro IV		>8 years	197,625
			Euro V		>8 years	104,828
			Euro 6		>8 years	
					4-8 years	
					<4 years	183,162
			Hybrid		All	30,036
		Electric			All	4,535
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	1,072
		Fuel-Injection	3-Ways		>8 years	8,237
			Euro III		>8 years	1,951
			Euro IV		>8 years	2,816
			Euro V		>8 years	363
			Euro 6		>8 years	1,494
					4-8 years	1,728
					<4 years	511
			Hybrid		All	2,114
		Electric			All	2,176

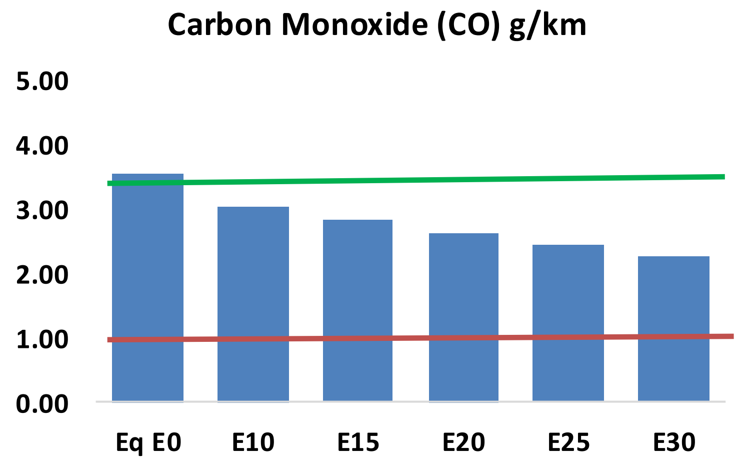
0 20.000 40.000 60.000 80.000 100.000 120.000 140.000 160.000 180.000 200.000

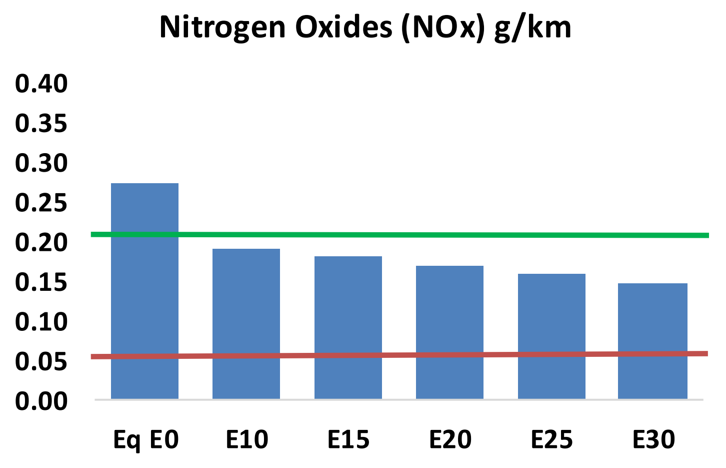
Vehicle Fleet: 2,358,574
Gasoline Vehicle Fleet: 923,614

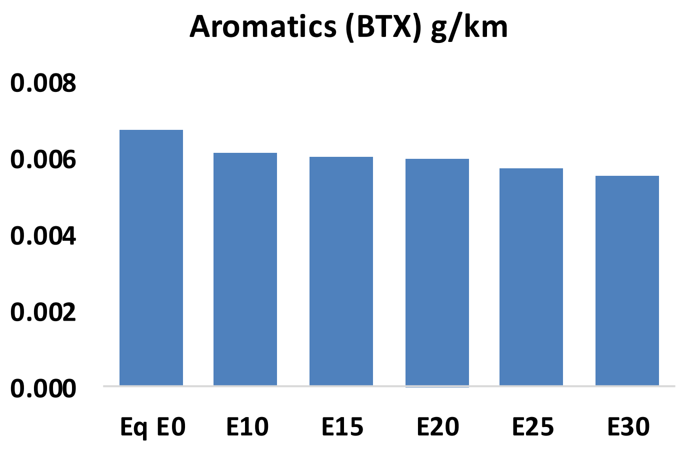
Source: UNECE, Eurostat, ACEA

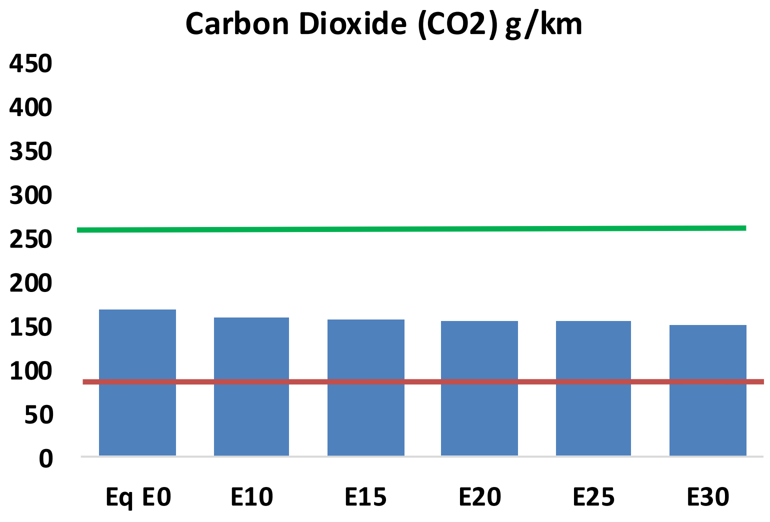
Croatia – Vehicular Emissions

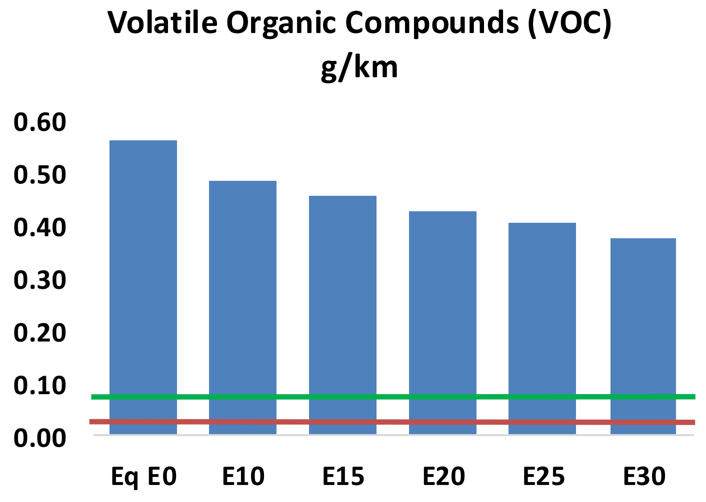
TIER III BIN 125 United States
Euro 6 Standard

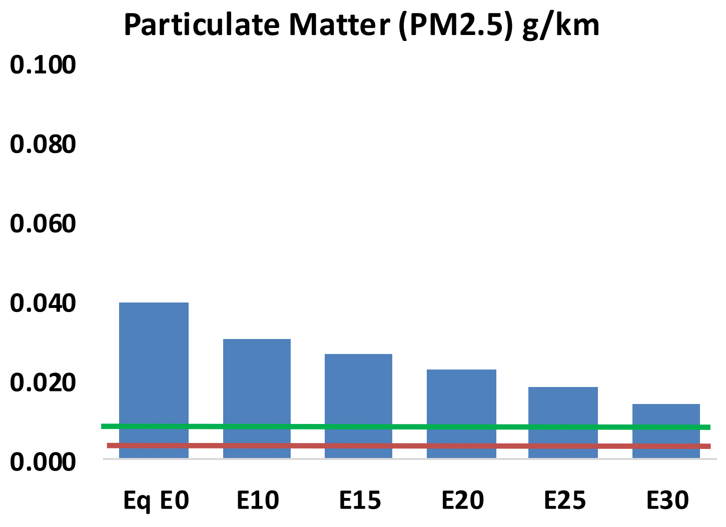












Source: Faro90

Croatia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	91,726	84,984	78,241	72,777	67,305	63,106	58,353	-15%	-27%
CO2	4,317,386	4,206,367	4,101,516	4,019,054	3,978,249	3,955,342	3,882,433	-5%	-8%
VOC	14,448	13,483	12,518	11,755	11,001	10,425	9,643	-13%	-24%
NOx	7,031	5,273	4,922	4,639	4,375	4,084	3,776	-30%	-38%
SOx	48	45	41	38	35	32	28	-15%	-28%
NH3	1,642	1,626	1,610	1,617	1,635	1,648	1,659	-2%	0%
N2O	59	59	59	59	60	61	62	-1%	2%
CH4	3,492	3,492	3,492	3,545	3,622	3,688	3,751	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	101	96	92	89	86	85	82	-9%	-14%
Acetaldehyde	293	336	493	751	1,023	1,169	1,381	68%	249%
Formaldehyde	1,099	1,069	1,246	1,463	1,533	1,696	1,846	13%	39%
Benzene	173	167	158	155	154	147	143	-9%	-11%
PM2.5	293	260	228	198	168	136	103	-22%	-43%
PM10	722	641	560	487	413	335	254	-22%	-43%

Source: Faro90

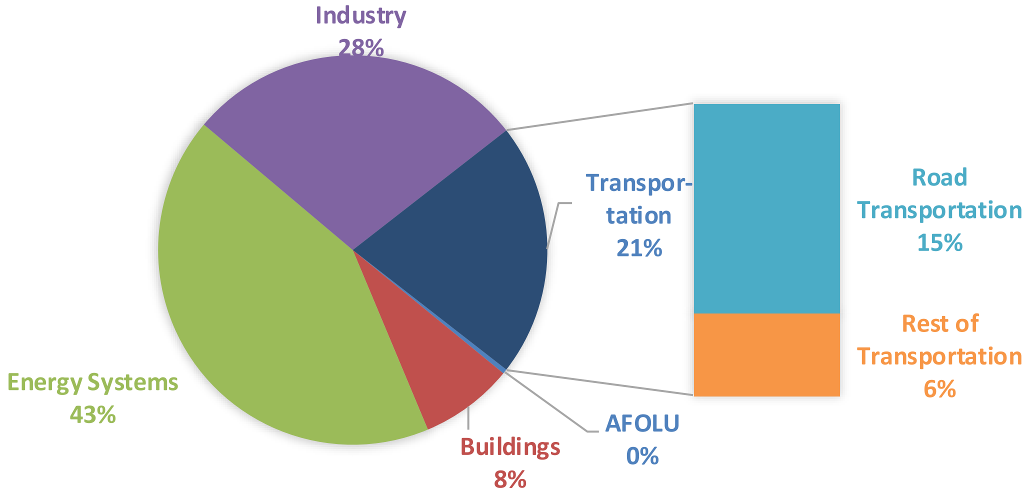
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

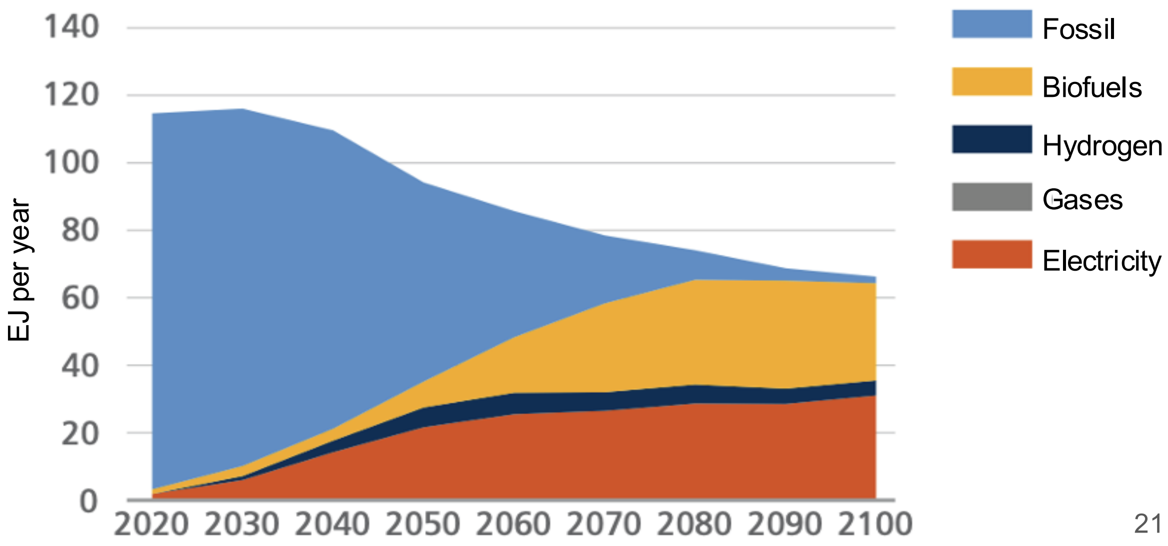
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

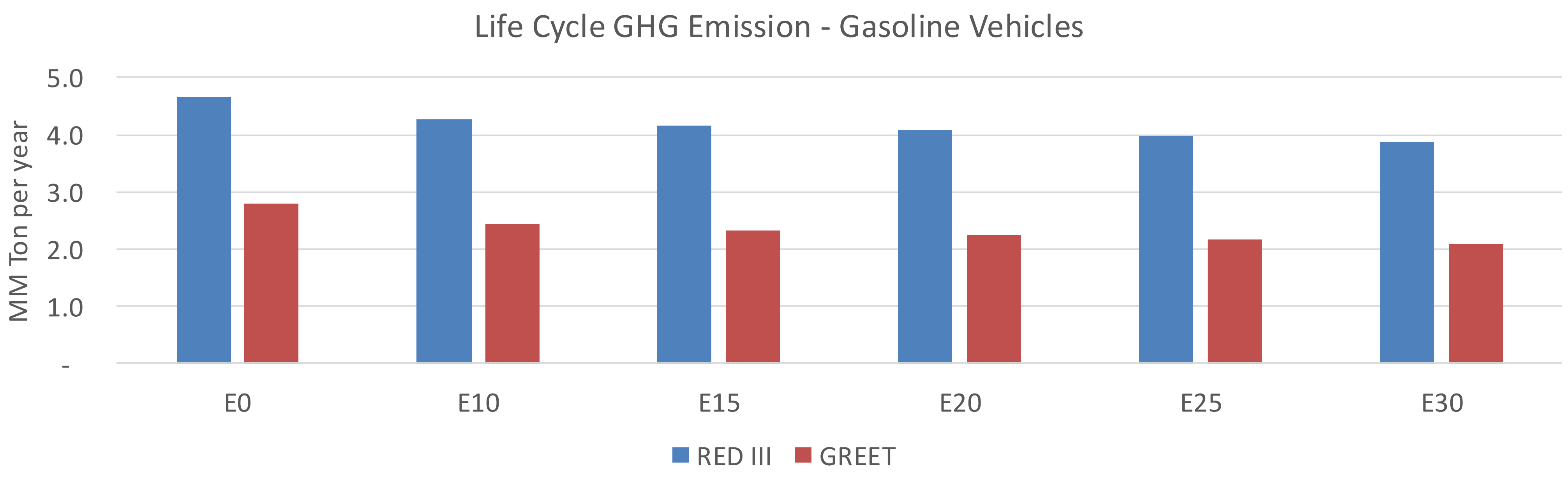
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Croatia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	17.9
Target @ 2035 (MMT/year)	10.1
Delta	7.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.3%	16.6%	19.4%	22.1%	25.6%
RED II/III	0.0%	8.5%	10.8%	12.7%	14.5%	16.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.7%	5.9%	6.9%	7.9%	9.1%
RED II/III	0.0%	5.1%	6.4%	7.6%	8.6%	10.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90